

DUNCAN CAMERON-STEINKE

Kelowna, British Columbia, Canada · (647) 936-0900 · dcameronsteinke@gmail.com

🌐 <https://duncancs.ca> · <https://linkedin.com/in/duncan-cameron-steinke/> · <https://github.com/DuncanCSt>

Quantitative researcher completing an MSc in Mathematics and Statistics, researching uncertainty-aware deep learning and regime-switching volatility forecasting in equity markets. Three years in industry to senior data engineer, plus two years co-founding an analytics startup.

RESEARCH EXPERIENCE

BELIEF DISTRIBUTION LEARNING

Mar 2026 – Present

Research paper (in preparation)

- Created a novel deep learning algorithm grounded in belief theory by combining Subjective Logic with Evidential Deep Learning.
- Introduced per-label uncertainty modeling using Dirichlet distributions.
- Evaluated against published Label Distribution Learning algorithms, demonstrating improved accuracy and better-calibrated uncertainty estimates.

Python

TensorFlow

Keras

NumPy

pandas

scikit-learn

Machine Learning

Deep Learning

Uncertainty Quantification

VOLATILITY FORECASTING WITH UNCERTAINTY-AWARE COVARIATES

Jul 2025 – Apr 2026

Research project

- Added forecasting to the hidden Markov model package hmmTMB.
- Coupled an HMM with an uncertainty-aware deep learning model using a custom loss function.
- Demonstrated improved forecasting of volatility regimes in the TSX composite index compared to HMM or deep learning alone.

Python

R

Hidden Markov Models

Time Series Forecasting

Deep Learning

FORECASTING RADIO RECOMBINATION LINE SCIENCE

Sep 2025 – Jul 2026

MSc thesis, UBC Okanagan

- Scientific research project predicting the performance of a next-generation radio telescope.
- Extrapolated science results given the telescope design and existing surveys as priors.
- Optimized dish geometry to improve signal resolution.

Python

Math

Statistics

Astronomy

Research

EXPERIENCE

SENIOR DATA ENGINEER

Aug 2022 – Apr 2024 1 yr 9 mos

ServiceTitan

- Owned the architecture, data modeling, and one direct report for an internal reporting platform built from scratch on Snowflake and Tableau, delivering metrics across billions of records to executive, business development, and customer success teams.
- Designed the data model and owned the pipelines behind a client-facing analytics product (Snowflake, SQL) serving 1,800+ customers.
- Designed the A/B testing methodology for an AI chat assistant automating call center support, using sample sizing and significance testing to measure impact on agent handling time.

ANALYTICS ENGINEER

Apr 2022 – Aug 2022 5 mos

Schedule Engine (acquired by ServiceTitan)

- Established the source of truth for reporting, defining business metrics where none existed and building the data analytics platform.

RESEARCH ENGINEER II, DATA SCIENCE

Jul 2021 – Feb 2022 8 mos

HomeX

- Deployed data ingestion microservices in Python and led client analytics projects.

CO-FOUNDER & CTO

Jun 2019 – Jul 2021 2 yrs 2 mos

Backr.ai (YouTube analytics startup)

- Co-founded a YouTube creator analytics platform, architecting a real-time streaming pipeline on GCP microservices in Python, with a React front end and a custom NLP model for comment classification and sentiment analysis.
- Won first prize (\$30K) at the DDQIC pitch competition and was selected for four accelerator programs including NextAI Montreal.
- Hired and led a team of three full-time developers and a data analyst, owning product roadmap and technical strategy from inception.
- Grew to 50 creator customers, including channels with 3M+ subscribers.

MACHINE LEARNING RESEARCH INTERN

Jan 2019 – Apr 2019 4 mos

Institute for Infocomm Research · Singapore

- Authored an NLP research paper (arXiv preprint) on unsupervised text expansion from pre-trained embeddings.
- Improved short-text classification accuracy under limited training data via generated pseudo-documents.

Research paper (arXiv) (<https://arxiv.org/abs/1909.07512>)

Singapore International Pre-Graduate Award

EDUCATION

THE UNIVERSITY OF BRITISH COLUMBIA

Sep 2024 – Aug 2026

Master of Science (MSc) · Kelowna, BC

Mathematics and Statistics

GPA: 4.0

- Coursework in non-parametric models, stochastic processes, differential equations, generalized linear models, and optimization.

QUEEN'S UNIVERSITY

Sep 2015 – May 2020

Bachelor of Applied Science (BASc) · Kingston, ON

Engineering Physics

GPA: 3.9

- Core studies in physics, mathematics, statistics, and computer programming.

Dean's Scholar · H. Jansen Memorial Scholarship (Awarded to the student with the highest standing in Physics I & II)

CERTIFICATIONS

MIT SCHWARZMAN COLLEGE OF COMPUTING

2022

Professional Certificate · Remote

Data Science and Machine Learning: Making Data-Driven Decisions

SKILLS

A/B Testing · Analytics · Astronomy · AWS · Business Intelligence · Data Engineering · Data Pipelines · Deep Learning · Entrepreneurship · Full Stack Development · Google Cloud · Hidden Markov Models · Keras · Leadership · Machine Learning · Math · Microservices · NLP · NumPy · pandas · Product Management · Production Software · Python · R · React · Research · scikit-learn · Snowflake · SQL · Statistics · Strategic Planning · Tableau · Teamwork · TensorFlow · Text Classification · Time Series Forecasting · Uncertainty Quantification